

MLFF Training-Data System Contract

Cross-cutting current-generation invariants

mdstats project

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Scope

This document is the cross-cutting current system contract for the mdstats MLFF training-data and fine-tuning workflow. The legacy filename is retained for stable references; this is **not** an implementation-stage plan.

It owns only invariants that span narrower specifications: evidence-role separation, dependency direction, identity/lineage, fitted-domain isolation, target-membership/target-size ownership, protocol/method identity, replay/monitor separation, sealed evaluation, calibration, bounded execution, and fail-closed current-generation publication.

Narrow specifications own exact module schemas, numerical constants, algorithms, storage formats, and runtime behavior. Current D1/D2 own scientific/numerical semantics; D3 architecture owns the higher-level dependency/ownership model. Workplans and historical documents are non-normative once binding semantics are promoted into current authority.

For restored post-selection P5, `mlff_post_selection_p5_spec.md` is the narrow current D4 owner. Broad DATA8-era `TrainingProtocolIdentity` contracts remain applicable only to separately current non-P5 consumers and historical provenance; they are not a second P5 authority.

Normative principles

1. Source facts, eligibility, evidence roles, fitted preparation, target membership, target size, weighting, exposure, checkpoint selection, validation, calibration, and acquisition are distinct record/decision families.

2. A frame that supplied a gradient is not independent validation evidence for that model.
3. Held-out cross-validation evaluates a frozen method/protocol and cannot control target size, fitting, stopping, or checkpoint choice for that fold.
4. After target selection, every **current restored-P5** fold has a gradient-training partition, a held-out outer-evaluation partition, and explicit independence/purge evidence. Its target checkpoint monitor is the external campaign-common `M_mon`, not selected-fold membership. Separately current non-P5 protocol families may retain their own accepted monitor topology.
5. Restored-P5 cross-validation trains a fresh model/optimizer lineage for each held-out fold and validates the complete `PostSelectionMethodIdentity` actually used by final production. A broad `TrainingProtocolIdentity` is not a second current P5 method identity.
6. Feature fitting, E0 fitting, label-derived difficulty evidence, and target-subset inputs inspect only the applicable authorized training partition; post-selection fold-local fits never inspect that fold's held-out labels. Foundation-P5 transfer validation may inspect governed geometry/composition classes from monitor/evaluation consumers but not their labels for fitting.
7. Current DATA6/DATA7 preparation publishes fitted inputs and evidence for the common target-size owner and SHALL NOT publish target membership or target size.
8. One canonical training order `pi_train` is the sole current target-membership authority; every candidate is the exact prefix `T_N = pi_train[:N]`.
9. The one target-size reducer is the sole owner of the *automatic* target-size diagnostic, which recommends a size and freezes nothing. The operator owns the provisional choice, restricted to the configured qualified candidate set. Monitor/replay/batch/pool cardinalities are different semantic types.
10. Exact target membership, the ordered collection of provisional target sizes, and their per-size effective role training horizons are frozen together, once, at `cross-validate` admission.
11. Target-size screening uses only authorized development/model-selection evidence. Held-out CV, calibration, and locked tests are forbidden inputs.
12. Locked tests cannot affect fitting, membership, size, protocol/method choice, stopping, checkpointing, calibration-policy choice, publication membership, or acquisition and are activated only after the applicable freeze boundary.
13. Replay training, replay monitoring, target monitoring, and target training preserve separate source/role identities.
14. Replay retention and mandatory physical/deployment integrity are hard admissibility constraints unless an explicit current scientific policy states otherwise.
15. Dynamic resampling/exposure semantics require an explicit current adapter/protocol; static files alone cannot claim them.
16. Calibration is bound to predictions from the actual frozen final committee/publication and an explicit applicability domain.
17. Active-learning child generations inherit prior evidence roles unless a new evaluation lineage explicitly reassigns them.
18. Retired campaign generations are rejected and re-prepared rather than migrated into current semantics. Retired derived target-size or P5 method state is detected before semantic reuse, candidate/checkpoint continuation, or descendant publication and is quarantined rather than translated. Only independently valid lower-level evidence whose recipes do not depend on retired semantics may be reused, and each is re-validated by its current owner.
19. Execution caches, worker scheduling, out-of-core layout, and other realization choices cannot change scientific/numerical identity or authoritative decisions.
20. Publication fails closed when required current-generation identities, upstream evidence, or schema/content validation are missing/incompatible.

21. Restored P5 has one campaign-common exact 256-frame target monitor `M_mon` from neutral label-usable `OUTER_MONITOR`, reused across every selected size, CV fold/seed, and final-production seed/run.
22. Common-monitor membership is sampled first under accepted D2; canonical P1 cross-role protected-relation qualification follows. Relation conflicts or exact-support shortfall fail closed rather than filtering/replacing/resampling members.
23. Foundation P5 uses selected-head foundation-residual E0 fitting plus composition-level transfer validation and does not bind the whole P3 `TargetSizeCommonTrainingPolicy` or inert P3 configuration-weight fitting.
24. Foundation P5 uses native MACE `UniversalLoss` through the one method-aware MACE seam. P3 target-size screening and P5 scratch keep their separately accepted weighted methods.
25. Retired target/replay **training-head scalar** weights are absent from restored P5; checkpoint/adaptive-stop target/replay score weights remain separately owned.
26. Foundation P5 binds replay/`pt_head` first then target before shuffle for multihead replay, no implicit target duplication, accepted `drop_last=true` geometry, and the qualified single-process path. Distributed foundation P5 fails closed pending separate D2-equivalence qualification.
27. Canonical single-source replay omission resolves `TRUE_DFT`; explicit pseudo-label training remains opt-in and requires independent `TRUE_DFT` replay monitoring. Ambiguous omitted legacy split-file semantics fail closed.
28. M3 remains P3 evidence and may support a separately authorized downstream probe, but it is not restored- P5 checkpoint, final-plan, final-seed-ranking, currentness, or publication ancestry.
29. `single_best_final_seed` consumes only already-frozen admissible final representatives and already-authenticated common-monitor target metric records under accepted target-only representative-ordering semantics; it performs no second target or M3 evaluation.

Core record ownership

Record / policy family		Owns		Must not own
TrainingDataSource	/ source records	source	bytes/controls/composition/label-domain lineage	frame eligibility or evidence role
TrainingFrameRecord		immutable	source-bound facts	eligibility, partition, membership, exposure
FrameEligibilityDecision		post-label/quality eligibility		partition or target membership
PartitionAssignment		one statistical role under DATA5/P1 policy		fitted quantities or target order
PartitionFeasibilityReport		whether requested evidence roles are supportable		fabricated independent evidence
PartitionIndependenceReport		actual independence	purge/duplicate limitations	stronger independence than observed
DATA6/7 fitted records		authorized training-partition descriptors/transforms/E0/difficulty/objective/weights/subset inputs		held-out labels, target membership, target size
target-size development split		one P_train/M3 split derived from neutral substrate		training order or size choice
canonical training order pi_train		one deterministic order whose prefixes are candidate subsets		evaluation populations or size choice
canonical evaluation ladder	pi_eval	nested direct	populations M1 subset M2 subset M3	training membership or size choice
common target-size preparation		one P3 preparation identity shared by every candidate size/optimizer seed		restored foundation-P5 method/preparation authority
target-size policy / reducer decision		configured candidate ladder, fidelity funnel, recommendation/no-recommendation		freezing size; P5 monitor construction; P5 CV
CampaignStore	provisional proposal	ordered (N, T_N identity, selection source, H_cv, H_prod)	collection of mutable	immutable ancestry; running screen work
CampaignStore frozen selection		selected sizes bound to exact T_N memberships plus role horizons		re-deciding size or accepting P5 method
OnlineTargetMonitorPolicy	/ common P5 monitor record	campaign-common target-monitor evidence membership		target-training size; held-out fold membership
ReplayMonitorPolicy		replay-monitor evidence set		target-training size or replay-training membership
PostSelectionMethodIdentity		complete current method identity	restored-P5	realized fold/monitor/fitted/checkpoint/evaluation descendants
P5 fitted-preparation record		authorized fit result plus foundation selected-head/transfer evidence where applicable		target-size ownership; inert P3 weighting authority
Campaign CV/final role plans, partition		role-specific P5 plan ancestry including same common monitor	alternates method/monitor authority	
Campaign CV/final role plans, partition		non-P5 consumers		

Every serialized current record SHALL carry a versioned schema, deterministic content identity, explicit upstream lineage, and explicit policy/failure identities as appropriate.

Identity and leakage contract

Source occurrence, geometry, and labels

Source occurrence (frame_uid or current equivalent), geometry fingerprint, label payload digest, and combined labeled-configuration fingerprint are distinct identities.

Geometry identity excludes energy/force/stress labels. Label identity includes selected labels and label-domain identity. Leakage audits use exact occurrence overlap, exact geometry overlap, exact labeled-configuration overlap, near-duplicate evidence where required, and forbidden temporal proximity.

Label-domain compatibility

Label-domain identity separates theory/electronic-structure identity, energy-reference identity, derivative/stress convention, numerical-quality profile, and software provenance. A current compatibility policy may accept non-semantic provenance differences but cannot silently merge incompatible theory or energy-reference domains.

One target MACE artifact family contains one compatible target label domain plus a separately identified replay lineage where replay is enabled.

Fitted-domain isolation

Raw physical/structural/event facts may be constructed before partitioning when the owning provider is partition-independent. Any learned/fitted transform—including scaling, PCA/whitening, fitted metrics, E0 corrections, or label-derived residual difficulty—is bound to a specific authorized gradient-training domain.

Before selection, the common P3 preparation is global. After admission of the ordered frozen collection, current restored-P5 fitted domains branch over each frozen size's exact membership T_N while sharing external M_{mon} :

```
P_train / common P3 target-size preparation
-> one pi_train and ordered collection of frozen sizes {N_selected}
-> one external campaign-common M_mon + P1 separation evidence
-> for each frozen size N_selected, exact  $T_N = pi\_train[:N\_selected]$ :
    |-> post-selection fold_training_partition_k(N)
    |   -> fold-local fitted P5 preparation from gradient-training labels only
    |   -> checkpoint/adaptive stop using external common M_mon
    |   -> held_out_evaluation_fold_k(N) only after checkpoint freeze
    |-> final-training fitted P5 preparation for complete  $T_N$ 
        -> fresh final production using same external M_mon
        -> publication before downstream qualification
```

For foundation P5, transfer validation additionally binds selected foundation checkpoint/head, exact residual-fit inputs/result, numerical rank/null-space evidence, and every governed target composition class. Monitor/held-out geometry/composition may participate in transfer feasibility; their labels do not enter the fit.

A reverse dependency from held-out evaluation into fitted products, target size, checkpoint selection, common-monitor membership, or final-publication ranking is prohibited.

Target membership and target-size contract

The current target-subset construction chain is:

```
neutral statistical substrate
-> one P_train / M3 development split
```

- > one canonical training order pi_train
- > one canonical evaluation ladder M1 subset M2 subset M3
- > one common target-size preparation
- > optional paired optimizer-seed automatic diagnostic (recommends only)
- > operator-owned provisional design (ordered collection of (N, H_cv, H_prod))
- > cross-validate admission
- > ordered collection of frozen entries (N_selected, T_selected = pi_train[:N_selected], and role horizons)

No current alternate, migration, or rescue branch exists. Retired derived target-size state is rejected before reuse rather than translated.

The configured candidate ladder, screen (n1,n2,n3), and independent production-horizon policy are owned by Architecture Part V and campaign configuration; they are not duplicated here.

Candidate membership at size N is exact prefix pi_train[:N]. cross-validate admission freezes the ordered collection of selected sizes, each with exact T_selected = pi_train[:N] and effective CV/production horizons. Identity projection stays role-specific: CV depends on the frozen CV horizon and never on production horizon, and vice versa.

Because every candidate is a prefix of one order, increasing N only adds frames; a non-monotone qualification result over nested increasing prefixes is an invariant failure.

Current campaign lifecycle

The public campaign lifecycle, including configuration initialization, is:

init -> doctor -> prepare -> select-target-size -> cross-validate -> train-production

storage is an orthogonal artifact-management command. status and advance project the same current owners. The campaign ends at fresh final-production closure; deployment, physical-observable, calibration, and locked-test qualification remain downstream contracts and cannot feed back into target-size or method selection.

Training-method/protocol and checkpoint contract

Restored current P5

PostSelectionMethodIdentity SHALL bind only current P5 method-bearing inputs, directly or through named component-policy digests. It SHALL NOT bind the whole P3 TargetSizeCommonTrainingPolicy merely because that object packages P3 objective/weighting/harness state.

Current P5 descendants additionally bind exact realized ancestry where appropriate:

- applicable selected target size and exact TargetBinding
- post-selection fold train/eval/purge identity where applicable
- replay source/training/monitor identities
- one exact common target-monitor record identity
- plan-level P1 monitor-vs-target separation evidence
- foundation residual-fit/transfer evidence where applicable
- checkpoint metric and replay-retention policy
- optimizer/LR/stopping/epoch policy
- seed policy
- precision/backend/runtime identity

A comparison or CV claim applies only to the current method/plan actually evaluated.

Checkpoint selection uses explicit target/focus/replay/property/integrity constraints. A candidate violating a mandatory constraint is inadmissible even if another target metric is lower. The held-out fold does not choose its checkpoint.

Broad DATA8/general protocols

TrainingProtocolIdentity may still bind foundation/model/head, selected membership, replay, objective/weights, exposure, checkpoint, optimizer, seed, precision/backend, and adapter/runtime identity for separately current non-P5 consumers that explicitly use that representation. It does not authorize restored P5.

MACE realization and exposure

Current MACE artifacts contain only supported labels/weights/masks/compact identities in interchange formats; extended provenance remains sidecar/content-addressed.

The adapter verifies current upstream behaviors on which the applicable method semantics depend: head ordering, loader realization, loss realization, checkpoint retention/control, precision/backend, and effective target/replay exposure.

Loss/exposure resolution is mode-specific:

P3 target-size screening	-> accepted weighted objective + complete-batch geometry
P5 scratch	-> separately accepted weighted method
P5 naive foundation adaptation	-> native UniversalLoss + accepted target-only foundation geometry
P5 multihead replay adaptation	-> native UniversalLoss + replay-first combined foundation geometry

There is no global loss-family switch that may change an unaffected method family. Intended exposure cannot substitute for realized exposure. Silent loader duplication or changed corpus order/counts fail closed unless the accepted method explicitly binds that behavior.

Foundation P5 remains on the accepted single-process path until a distributed realization is separately D2-qualified.

Runtime/package locks are owned by narrow current runtime specifications and may evolve independently only when applicable equivalence/currentness obligations are satisfied.

Sealed evaluation, calibration, and active learning

Development, calibration, and locked evaluation artifacts remain role-separated. Locked-test configuration/path access is absent from development control flow until explicit activation after the applicable protocol/publication freeze.

Calibration numerical thresholds derive from the actual frozen final committee/publication and a dedicated authorized calibration cohort. Applicability/transfer decisions explicitly distinguish within-domain, rank-only, recalibration-required, and incompatible-domain behavior.

Active-learning labels create a new development generation. Existing role assignments are inherited by default; repartitioning previously classified evidence creates a new evaluation lineage.

Bounded execution and persistence

Scientific policy must be realizable without duplicating product-scale state per target-size rung. The current architectural materialization remains one fitted-input authority, one canonical training order, one common P3 target-size preparation, prefix metadata for candidate rungs, and only currently authorized training artifacts. Restored P5 adds one shared common-monitor record rather than one monitor per fold/size/seed.

Persistent execution caches are reconstructible unless another current specification explicitly makes them scientific evidence. Every cache validates semantic inputs and payload integrity. Corrupt/stale state is rebuilt or fails cleanly; it never changes policy to rescue a run.

Worker count, queue ordering, chunking, file-backed versus in-memory layout, and cache path are non-semantic only under the applicable exact-equivalence contract.

Current-generation publication and failure rules

Current products SHALL fail closed when required source/label identity, evidence roles, fitted-partition lineage, target-size decision, replay/monitor lineage, post-selection acceptance, current method/protocol identity, runtime behavior, or publication payload validation is missing/incompatible.

For restored P5, incompatible historical weighted-stress foundation trajectories, fold-local target-monitor plans, M3-dependent P5 final plans/publications, from-scratch-E0 foundation preparations, target-first replay exposure records, missing-transfer preparations, and broad DATA8 TrainingProtocolIdentity payloads used as P5 authorization remain historical and cannot become current through deserialization.

Unsupported historical campaign schemas are not current compatibility obligations. Current code may retain low-level readers for forensic purposes, but those readers cannot create a second product-semantic path. Independently valid lower-level evidence may be reused only through its current owner and validation rules.

Extension rule

A new feature/provider may enrich raw or DATA7 inputs without creating another membership selector. A new selector objective, target-size population, evidence role, loss function, stopping rule, common-monitor method, P5 method owner, final-publication selection rule, or compatibility generation changes accepted semantics at its owning D1-D4 layer and requires explicit revision plus qualification. Lower-layer implementation cannot silently compensate for an upstream change.